

Data Analytics (DA)

Module-End -3

Module End Assignment:

“Analyzing E-Learning Platform Purchases using MySQL”

Assignment Task Overview

Problem Statement:

An online learning platform sells various digital courses to learners across different countries. Each learner can purchase multiple courses, and each course belongs to a specific category.

The management team wants to analyze purchase data to understand:

- Sales trends
- Learner behavior
- Popular course categories

Your task is to use MySQL to design, query, and extend a relational database to derive meaningful insights from this e-learning purchase data.

Dataset Description:

You will work with the following three tables:

Table: *learners*

<i>Column</i>	<i>Description</i>
<i>learner_id</i>	<i>Primary Key</i>
<i>full_name</i>	<i>Learner name</i>
<i>country</i>	<i>Country of residence</i>

Table: *courses*

<i>Column</i>	<i>Description</i>
<i>course_id</i>	<i>Primary Key</i>
<i>course_name</i>	<i>Course title</i>
<i>category</i>	<i>Course category</i>
<i>unit_price</i>	<i>Price per course</i>

Table: *purchases*

<i>Columns</i>	<i>Description</i>
<i>purchase_id</i>	<i>Primary Key</i>
<i>learner_id</i>	<i>Foreign Key → learners</i>
<i>course_id</i>	<i>Foreign Key → courses</i>
<i>quantity</i>	<i>Number of courses purchased</i>
<i>purchase_date</i>	<i>Date of purchase</i>

Task

1. Database Setup & Data Entry

- Create a new database.
- Create the three tables with proper:
 - Primary keys, Foreign keys, Appropriate data types
- Insert sample data:
 - 4–5 learners
 - 4–5 courses (different categories)
 - 6–8 purchase records

2. Data Exploration Using Joins

Use *INNER*, *LEFT*, and *RIGHT JOIN* to:

- Combine learner, course, and purchase data
- *Display*: Learner name, Course name, Category, Quantity, Total amount, Purchase date
- Presentation Requirements:
 - Format currency to 2 decimal places
 - Use column aliases
 - Sort by the highest total amount

3. Core Analytical Queries (Q1–Q5)

Q1. Display each learner's total spending with their country.

Q2. Find the top 3 most purchased courses by quantity.

Q3. Show each category's:

- Total revenue
- Number of unique learners

Q4. List learners who purchased from more than one category.

Q5. Identify courses never purchased.

4. Subqueries & Correlated Subqueries

Q6. Find learners whose total spending is above the average learner spending.

Q7. Display courses whose price is higher than any course in the 'Beginner' category.

Q8 . Find learners who spent more than the average spending in their country.

5. CTE, CASE, View, and NULL Handling

Q9. Use a CTE to calculate total spending per learner, then:

Display learners with spending above 10,000.

Q10. CASE Expression

Classify learners based on spending:

- Above 15,000 → "High Value",
- 8,000–15,000 → "Medium Value",
- Below 8,000 → "Low Value".

Q11 . NULL Handling

- Display all courses and replace NULL purchase counts with 0 using: `IFNULL()` or `COALESCE()`

Q12 . View

- Create a view: `category_performance_view`
- Showing:
 - *Category*
 - *Total revenue*
 - *Number of purchases*
 - *Average revenue per purchase*

Deliverables

Submit a project folder containing:

1. MySQL Script (.sql) that includes:

- Database and table creation statements
- Sample data inserts
- Join queries (INNER, LEFT, RIGHT)
- Analytical queries
- Subqueries (single, multi-row, correlated)
- CTE, CASE, and NULL-handling queries
- view

2. One-page Summary Report (Word or PDF) including:

- Key insights from the analysis
- Brief recommendations (e.g., “Focus marketing on top-performing course categories”)

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Proficiency
Database Setup & Data Entry	SQL Environment Management	Creating databases, tables, defining data types, inserting seed data	Beginner
Data Exploration with Joins	SQL Querying	INNER/LEFT/RIGHT joins, multi-table queries	Intermediate
Aggregation & Filtering	Data Analysis using SQL	GROUP BY, HAVING, WHERE, ORDER BY, LIMIT	Intermediate
Derived Columns & Calculations	Analytical Logic	Using expressions for revenue, totals, and ratios	Intermediate
Subqueries	Advanced SQL	Single, multi-row, correlated	Intermediate
CTE & CASE	Query Structuring	WITH clause, conditional logic	Intermediate
NULL Handling	Data Cleaning	IFNULL, COALESCE	Beginner
Views	Data Abstraction	Simple & complex views	Intermediate
Descriptive Reporting	Data Summarisation	Summarizing learner and course-level metrics	Intermediate
Result Presentation & Interpretation	Communication	Clear query output formatting, readable column aliases, and short insights	Beginner

Rubric & Mark Allocation (Total 25 Marks)

Skill / Component	Full Marks	Criteria for Full Marks	Partial Credit Description	Partial Marks Range
Database Setup & Data Entry	3	Database, tables, keys created correctly; sample data inserted	Minor syntax or constraint issues	1–2
Joins & Data Exploration	5	Correct INNER, LEFT, and RIGHT joins with clear output and aliases	Some joins incorrect or incompletely	2–4
Aggregation & Filtering	4	Accurate use of GROUP BY, WHERE, HAVING, ORDER BY	Minor grouping or filtering errors	2–4
Analytical Queries	6	All business questions answered correctly with valid logic	One or two queries are incorrect/missing	3–5
Advanced SQL (Subqueries, CTE, CASE, NULL Handling)	3	Correct use of subqueries, CTE, CASE, IFNULL/COALESCE	Partial or inconsistent logic	1–2
Views	2	views created correctly	One view incorrect or missing	1
Report Summary & Recommendations	2	Clear insights and actionable recommendations	Vague or minimal explanation	1

DA Skill Taxonomy Structure

- **Database Setup:** Create MySQL databases, tables, and relationships
- **Data Insertion:** Populate tables with valid, consistent data
- **Joins & Exploration:** Use INNER, LEFT, and RIGHT joins to combine tables
- **Aggregation & Filtering:** Apply GROUP BY, WHERE, HAVING, ORDER BY
- **Calculations:** Compute revenue, totals, and counts
- **Common Table Expressions and views**
- **Analytical SQL:** Answer business questions using SQL logic
- **Result Presentation:** Format outputs with clear aliases
- **Insights:** Summarize findings and give recommendations

Evaluation Guidelines

- **Open and review the submitted .sql script** to ensure:
 - Tables are correctly created and populated.
 - All data exploration using joins and five analytical queries are present and executable.
- **Check the correctness of joins, grouping, and calculations.**
- **For each section,** match student work to the specific skill/sub-skill area in the taxonomy.
- **Assign full or partial marks** according to the rubric.
 - Full marks: accurate syntax, logic, and clean formatting.
 - Partial marks: minor issues, incomplete queries, or inconsistent naming.
- **Comment** on strengths and improvement points for each section.
- **Maintain objectivity**—assess the correctness, clarity, and relevance of SQL queries.

Achievement Criteria & Passing Rules

- At least partial (non-zero) marks must be obtained in **every assignment section**.
- Students must fully achieve at least **two major skills: database setup & design, and SQL data analysis (joins, aggregations, calculations, CTEs, and views)**.
- All required SQL scripts must be **executable and logically correct**.
- **Feedback and a detailed scorecard** must be provided for every evaluated submission.